

Daily Sequences Drill #5 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Monotonicity & Boundedness · Fixed Points · Non-Modular Periodicity · Floor/Ceiling Sequences.

Section A Rapid Recognition

1. **Answer:** Decreasing

Method: $a_{n+1} - a_n = (5 - (n + 1)) - (5 - n) = -1 < 0$ for every n .

Investigate further: Every AP with negative common difference is decreasing — this is just Day 1's definition, restated in “increasing/decreasing” language.

2. **Answer:** Not bounded above; bounded below (by 1).

Method: $n^2 \rightarrow \infty$ as $n \rightarrow \infty$, so no upper bound exists. But $n^2 \geq 1$ for every positive integer n , so 1 is a lower bound.

Investigate further: A sequence can be bounded on one side only — always check both directions separately.

3. **Answer:** $x = 3$

Method: $x = 2x - 3 \Rightarrow -x = -3 \Rightarrow x = 3$.

Investigate further: Check: $f(3) = 2(3) - 3 = 3$ ✓ — once a sequence generated by f lands exactly on 3, every later term stays at 3 forever.

4. **Answer:** $a_2 = \frac{1}{2}, a_3 = 2, a_4 = \frac{1}{2}$

Method: $a_2 = \frac{1}{a_1} = \frac{1}{2}$. $a_3 = \frac{1}{a_2} = 2$. $a_4 = \frac{1}{a_3} = \frac{1}{2}$.

Investigate further: This sequence has period 2: odd-indexed terms are 2, even-indexed terms are $\frac{1}{2}$, forever — an early taste of today's periodicity theme.

5. **Answer:** $\lfloor 3.7 \rfloor = 3, \lceil 3.2 \rceil = 4$

Method: Floor rounds down to the nearest integer; ceiling rounds up.

Investigate further: Both operations leave an already-integer value unchanged — $\lfloor 5 \rfloor = \lceil 5 \rceil = 5$.

6. **Answer:** Yes, converges to 0.

Method: Both $\left(\frac{1}{2}\right)^n \rightarrow 0$ and $\left(\frac{1}{3}\right)^n \rightarrow 0$ as $n \rightarrow \infty$, since both bases have magnitude less than 1. Their sum also $\rightarrow 0$.

Investigate further: This is literally Day 4's D5 result, applied to specific numbers: $a_n = Ar_1^n + Br_2^n$ with $A = B = 1$, $r_1 = \frac{1}{2}$, $r_2 = \frac{1}{3}$, both inside the unit circle.

7. **Answer:** Bounded (yes); does not converge.

Method: Every term is either 1 or -1 , so $-1 \leq a_n \leq 1$ always (bounded). But the sequence never settles near a single value — it alternates forever.

Investigate further: This is the standard counterexample to “boundedness implies convergence” — see A8 and Common Trap 1.

8. **Answer:** False

Method: $a_n = (-1)^n$ (A7) is a direct counterexample: bounded, but not convergent.

Investigate further: The correct combined statement is: bounded *and monotonic* together imply convergence — boundedness alone is not enough.

9. **Answer:** Increasing

Method: $a_1 = \frac{1}{2}$, $a_2 = \frac{2}{3}$, $a_3 = \frac{3}{4}$ — each term is bigger than the last.

Investigate further: B1 proves this rigorously for all n , not just the first three terms.

10. **Answer:** 16

Method: $\lfloor \sqrt{n} \rfloor = 4$ needs $4 \leq \sqrt{n} < 5$, i.e. $16 \leq n < 25$. The smallest such n is 16.

Investigate further: This is the boundary case of C5's counting question — always convert a floor equation into a double inequality on n before searching.

Section B Manipulation Drills

1. **Answer:** Proof: $a_{n+1} - a_n = \frac{1}{(n+1)(n+2)} > 0$.

Method: $a_{n+1} - a_n = \frac{n+1}{n+2} - \frac{n}{n+1} = \frac{(n+1)^2 - n(n+2)}{(n+2)(n+1)} = \frac{n^2 + 2n + 1 - n^2 - 2n}{(n+2)(n+1)} = \frac{1}{(n+1)(n+2)}$, which is positive for every $n \geq 1$. $\square$

Investigate further: This “common denominator, then simplify the numerator” technique for proving $a_{n+1} - a_n > 0$ works on almost any rational sequence — practice it on B2's boundedness question next.

2. **Answer:** A) Yes, supremum 1.

Method: $a_n = \frac{n}{n+1} = 1 - \frac{1}{n+1} < 1$ for every n , and $a_n \rightarrow 1$ as $n \rightarrow \infty$ — so 1 is the smallest possible upper bound (the supremum), even though no term ever equals it exactly.

Investigate further: Rewriting $\frac{n}{n+1}$ as $1 - \frac{1}{n+1}$ makes both the increasing behaviour (B1) and the bound (here) obvious at a glance — always look for this kind of rewrite on a rational sequence.

3. **Answer:** B) $\frac{1}{3}$

Method: The sequence has period 2: $a_1 = 3, a_2 = \frac{1}{3}, a_3 = 3, a_4 = \frac{1}{3}, \dots$ — odd indices give 3, even indices give $\frac{1}{3}$. Since 100 is even, $a_{100} = \frac{1}{3}$.

Investigate further: Always reduce a large index modulo the period before doing anything else — $100 \bmod 2 = 0$, matching the “even” case.

4. **Answer:** $x = \pm 1$

Method: $\frac{1}{x} = x \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$.

Investigate further: A function can have more than one fixed point — $f(x) = \frac{1}{x}$ has two, unlike A3's $f(x) = 2x - 3$, which has exactly one (any non-identity linear function has at most one).

5. **Answer:** A) $a_n = 5$ for all n .

Method: $a_1 = 5$ is exactly the fixed point, so $a_2 = f(5) = 5$, and by the same reasoning every later term stays at 5 forever.

Investigate further: This is the simplest possible case of C8's general theorem: if a sequence *starts* at a fixed point, it never leaves.

6. **Answer:** C) $\frac{1}{n}$

Method: $\frac{1}{n}$ strictly decreases as n increases. A) and B) oscillate without settling into one direction; D) cycles through 1, 2, 0, 1, 2, 0, $\dots$ forever.

Investigate further: A sequence is monotonic only if *every* consecutive pair moves the same direction — one exception anywhere disqualifies it, which is why periodic sequences (like D) are essentially never monotonic unless constant.

7. **Answer:** B) -3

Method: Floor always rounds toward $-\infty$, not toward 0: the greatest integer that is still ≤ -2.3 is -3 , not -2 .

Investigate further: This is the single most common floor-function slip — for negative numbers, “round down” means *more* negative, not closer to zero.

8. **Answer:** A) 0, 1, 1, 2, 2

Method: $a_1 = \lfloor 0.5 \rfloor = 0, a_2 = \lfloor 1 \rfloor = 1, a_3 = \lfloor 1.5 \rfloor = 1, a_4 = \lfloor 2 \rfloor = 2, a_5 = \lfloor 2.5 \rfloor = 2$.

Investigate further: This sequence repeats each integer value exactly twice (except possibly the first) — a simple example of how floor division “compresses” a sequence.

9. **Answer:** Converges to 0.

Method: $\left| \frac{(-1)^n}{n} \right| = \frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$, and since $-\frac{1}{n} \leq a_n \leq \frac{1}{n}$, the sequence is squeezed to 0 regardless of the alternating sign.

Investigate further: This is a genuinely different situation from A7's $(-1)^n$: here the *magnitude* shrinks to 0 as well as the sign alternating, so the oscillation itself dies out and convergence does

| happen.

10. **Answer:** A) $d > 0$

Method: $a_{n+1} - a_n = d$ for every n in an AP — strictly increasing means this difference must be positive, i.e. $d > 0$.

| **Investigate further:** Compare to Day 1: this is the same fact as Day 1's toolkit, now phrased in this sheet's "increasing/decreasing" language rather than "common difference."

Section C Substitution & Structure

1. **Answer:** A) 2

Method: $a_1 = 2$. $a_2 = \frac{2-1}{2+1} = \frac{1}{3}$. $a_3 = \frac{1/3-1}{1/3+1} = \frac{-2/3}{4/3} = -\frac{1}{2}$. $a_4 = \frac{-1/2-1}{-1/2+1} = \frac{-3/2}{1/2} = -3$.
 $a_5 = \frac{-3-1}{-3+1} = \frac{-4}{-2} = 2$.

| **Investigate further:** This Möbius map has exact period 4: $a_5 = a_1$, so the cycle $2, \frac{1}{3}, -\frac{1}{2}, -3$ repeats forever — a classical fact about the transformation $x \mapsto \frac{x-1}{x+1}$.

2. **Answer:** A) 2

Method: The sequence has period 4 (from C1). $101 = 4(25) + 1$, so $101 \equiv 1 \pmod{4}$, meaning $a_{101} = a_1 = 2$.

| **Investigate further:** Exactly the same index-reduction technique as B3, just with period 4 instead of 2 — always find the period first, then reduce the target index modulo it.

3. **Answer:** A) Increasing and bounded above by 2; converges to 2.

Method: Computing terms: $a_1 = 1, a_2 = \sqrt{3} \approx 1.73, a_3 \approx 1.93, a_4 \approx 1.98, \dots$ — visibly increasing and approaching 2. (D1 proves this rigorously.)

| **Investigate further:** The fixed point of $f(x) = \sqrt{x+2}$ solves $x = \sqrt{x+2} \Rightarrow x^2 = x+2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x-2)(x+1) = 0 \Rightarrow x = 2$ (rejecting $x = -1$ since $f(x) \geq 0$ always) — exactly matching the numerically observed limit.

4. **Answer:** C) 5

Method: Check the primes in order: $n = 2$: $w_2 = 15 = 3 \times 5$, a multiple of both — not a counterexample. $n = 3$: $w_3 = 18 = 3 \times 6$, a multiple of 3 — not a counterexample. $n = 5$: $w_5 = 31$, which is neither a multiple of 3 ($31 = 3 \times 10 + 1$) nor of 5 — this IS a counterexample. Since 5 is the smallest prime that fails, it's the answer.

| **Investigate further:** The real TMUA question (2021 Paper 2 Q10) uses the sequence 15, 21, 30, 37, 44, 51, 59 and the claim "multiple of 3 or 5," also landing on $n = 5$ as the smallest counterexample — always check primes in increasing order and stop at the first failure, rather than checking every term first.

5. **Answer:** C) 15

Method: $a_n = 7$ needs $7 \leq \sqrt{n} < 8$, i.e. $49 \leq n < 64$. The integers from 49 to 63 inclusive number $63 - 49 + 1 = 15$.

Investigate further: This is A10's boundary-finding technique extended to counting a whole range — always convert to a double inequality on n first, then count the integers in that range.

6. **Answer:** A) Periodic, period 3.

Method: $a_n = n - 3\lfloor n/3 \rfloor$ is exactly the remainder of n divided by 3: for $n = 1, 2, 3, 4, 5, 6, \dots$ this gives $1, 2, 0, 1, 2, 0, \dots$ — repeating every 3 terms.

Investigate further: Floor notation and the mod operator describe exactly the same thing here — $n \bmod 3$ and $n - 3\lfloor n/3 \rfloor$ are identical formulas written two different ways.

7. **Answer:** A) Always increasing (or constant), for any starting value.

Method: $a_{n+1} - a_n = (a_n^2 - a_n + 1) - a_n = a_n^2 - 2a_n + 1 = (a_n - 1)^2 \geq 0$ always, regardless of a_n 's value — so $a_{n+1} \geq a_n$ for every n , for *any* real starting value, not just $0 < a_1 < 1$.

Investigate further: The restriction " $0 < a_1 < 1$ " in the question is a red herring — the algebra shows the result holds unconditionally. Recognising when a stated constraint isn't actually load-bearing is itself a skill (see Common Trap 5).

8. **Answer:** A) $f(L) = L$ — L must be a fixed point of f .

Method: Taking the limit of both sides of $a_{n+1} = f(a_n)$ as $n \rightarrow \infty$: the left side $\rightarrow L$, and since f is continuous, the right side $\rightarrow f(L)$. So $L = f(L)$.

Investigate further: This is the theorem underlying every fixed-point computation in this sheet (A3, B4, B5, C3, D1, D4) — it's why solving $f(x) = x$ is always the right first move when a recursively-defined sequence is claimed (or suspected) to converge.

Section D Challenge

1. **Answer:** Bounded above by 2, strictly increasing, converges to 2.

Method: *Bounded above by 2 (induction):* Base case $a_1 = 1 < 2$. Inductive step: assume $a_k < 2$; then $a_{k+1} = \sqrt{a_k + 2} < \sqrt{2 + 2} = 2$. By induction, $a_n < 2$ for all n .

Strictly increasing: Since $0 < a_n < 2$, we have $(a_n - 2) < 0$ and $(a_n + 1) > 0$, so $(a_n - 2)(a_n + 1) < 0$, i.e. $a_n^2 - a_n - 2 < 0$, i.e. $a_n + 2 > a_n^2$. Since $a_n \geq 0$, taking square roots preserves the inequality: $\sqrt{a_n + 2} > a_n$, i.e. $a_{n+1} > a_n$.

Convergence: An increasing sequence bounded above must converge (Monotone Convergence Theorem). Taking the limit L of $a_{n+1} = \sqrt{a_n + 2}$: $L = \sqrt{L + 2} \Rightarrow L^2 = L + 2 \Rightarrow (L - 2)(L + 1) = 0 \Rightarrow L = 2$ or $L = -1$. Since every $a_n > 0$, $L \geq 0$, so $L = 2$. $\square$

Investigate further: This is the complete, rigorous version of what C3 only observed numerically — notice that all three pieces (boundedness, monotonicity, and the limit equation) reuse ideas from earlier in the sheet: induction, the $(a_n - 2)(a_n + 1)$ factoring trick from C7's style of algebra, and C8's fixed-point theorem.

2. **Answer:** B) 3

Method: $a_1 = 2$. $a_2 = \frac{1}{1-2} = -1$. $a_3 = \frac{1}{1-(-1)} = \frac{1}{2}$. $a_4 = \frac{1}{1-\frac{1}{2}} = 2 = a_1$.

Investigate further: This is a different Möbius map from C1/C2's — $f(x) = \frac{1}{1-x}$ has order 3 under composition (a classical fact), while C1's $f(x) = \frac{x-1}{x+1}$ has order 4. Both are genuine, exact cycles — always iterate directly rather than assuming a map is aperiodic.

3. Answer: C) 8

Method: $a_1 = 100, a_2 = 50, a_3 = 25, a_4 = 12, a_5 = 6, a_6 = 3, a_7 = 1, a_8 = 0$.

Investigate further: Repeated floor-halving reaches 0 in roughly $\log_2(a_1)$ steps — here $\log_2 100 \approx 6.6$, and the actual count (7 steps from 100 to 0, i.e. $a_8 = 0$) matches that estimate closely, since each floor operation can only lose at most one extra step compared to plain halving.

4. Answer: A) Decreasing (after the first term) and bounded below by 2; converges rapidly to 2.

Method: $a_1 = 3, a_2 = \frac{3+4/3}{2} = \frac{13/3}{2} = \frac{13}{6} \approx 2.167, a_3 \approx 2.006, a_4 \approx 2.00002$ — rapidly approaching 2 from above. The fixed point solves $x = \frac{x+4/x}{2} \Rightarrow 2x = x + \frac{4}{x} \Rightarrow x = \frac{4}{x} \Rightarrow x^2 = 4 \Rightarrow x = 2$ (taking the positive root, since all terms stay positive).

Investigate further: This is Newton's method (the Babylonian method) for computing $\sqrt{4}$ — the same iteration $a_{n+1} = \frac{1}{2} \left(a_n + \frac{c}{a_n} \right)$ computes $\sqrt{c}$ for any $c > 0$, converging extremely fast (the number of correct digits roughly doubles every step).

5. Answer: A) Decreasing for all n .

Method: $a_2 = \frac{1}{2} \left(1 - \frac{1}{2} \right) = \frac{1}{4}, a_3 = \frac{1}{4} \left(1 - \frac{1}{4} \right) = \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16}$. In general, $a_{n+1} - a_n = a_n(1 - a_n) - a_n = a_n(1 - a_n - 1) = -a_n^2 \leq 0$ always, so the sequence never increases.

Investigate further: This is C7's exact technique in reverse-flavour: there, $(a_n - 1)^2 \geq 0$ forced the sequence up; here, $-a_n^2 \leq 0$ forces it down. Both times, rewriting $a_{n+1} - a_n$ as $\pm(\text{something})^2$ settles monotonicity instantly, without needing to compute many terms or invoke boundedness at all.

Top 5 Patterns — Worksheet #5

- Monotonicity via the sign of $a_{n+1} - a_n$.** Often collapses to $\pm(\text{something})^2$, settling the direction instantly without computing terms. *(see A1, A9, B1, C7, D5)*
- Fixed points: solve $f(L) = L$.** If a sequence converges, its limit must be a fixed point of the generating function. *(see A3, B4, B5, C8, D4)*
- Boundedness + monotonicity together imply convergence.** Neither alone is enough — see Common Trap 1. *(see C3, D1)*
- Möbius maps can have small, exact periods under repeated composition.** Check by direct iteration — don't assume a map is aperiodic just because it looks complicated. *(see C1, C2, D2)*
- Floor/ceiling notation is just an explicit “round down/up.”** Expressions like $\lfloor n/k \rfloor$ often secretly encode periodicity (a remainder) or a counting bound. *(see A5, A10, B7, B8, C5, C6, D3)*

Common Traps — Worksheet #5

1. **Assuming boundedness alone implies convergence.** $(-1)^n$ is bounded but never converges — monotonicity is also needed. (see A7, A8)
2. **Forgetting floor rounds DOWN even for negative numbers.** $\lfloor -2.3 \rfloor = -3$, not -2 . (see B7)
3. **Assuming a sequence must be eventually constant or simply diverging.** Some sequences are genuinely, exactly periodic with period > 1 forever. (see C1, C2, D2)
4. **Trusting a few computed terms as proof of monotonicity.** Numerical evidence is not a proof — always confirm the sign of $a_{n+1} - a_n$ algebraically for all n . (see D1)
5. **Assuming a stated restriction on initial conditions is load-bearing.** Some monotonicity results hold far more generally than a question implies — check whether the algebra actually needs the stated constraint. (see C7)

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths